



SmartDaily: Providing all-inclusive smart community services with Google Cloud

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SmartDaily deploys its smart community services in Google Cloud to provide wide-ranging community management and domestic services with high stability, while leveraging data analytics to

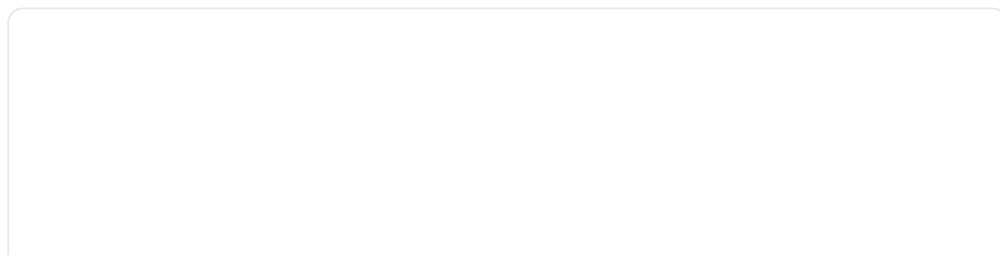
optimize its marketing performance, service quality, and operations.

Close to [80 percent](#) of Taiwan's population lives in urban areas, with many of them residing in apartment complexes accommodating dozens to hundreds of households. These apartment buildings are often managed by a committee formed within the community, who would look for ways to solve all kinds of community life issues for residents.

To support these initiatives, KNST CO., LTD. was founded in 1997 to offer a wide range of smart community services. The company started as a broadband internet provider and has been in close contact with many apartment complex-based communities across Taiwan. As smart home solutions became increasingly popular, KNST CO., LTD. launched [SmartDaily](#), a one-stop smart community service platform to enhance community management efficiency and living experiences in 2016. Providing various Artificial Intelligence of Things (AIoT)-powered community management solutions like smart access control and domestic services like air conditioning cleaning, SmartDaily currently serves nearly 9,000 communities and more than million families.

"Services offered by the community committees play an important role in the living experiences of people who reside in apartment complexes," explains Nathan Chang, R&D Director at SmartDaily. "By digitalizing community management and domestic services, we provide a range of conveniences for residents. Users can safely pay management fees, book appliance cleaning, home cleaning, and other home services on our platform."

At first, SmartDaily deployed its smart community service platform in an on-prem environment. However, as the traffic of its platform grew, the team frequently encountered technical issues like electricity shortage, hardware errors, and insufficient computing resources that disrupted its services. To improve its service quality and data security, the company migrated to [Google Cloud](#) in 2017, as it was the only cloud provider that had a data center in Taiwan at that time and offered the most cost-effective cloud infrastructure.





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**Supporting a secure, all-inclusive smart
community service platform**

With technical advice given by Google Cloud partner [CloudMile](#), KNST CO., LTD. smoothly migrated SmartDaily from its on-prem

servers to Google Cloud. The support by CloudMile continues to this day as the team provides timely support whenever the SmartDaily team needs to optimize its IT architecture or develop new features. For example, it suggested SmartDaily to replace virtual machines (VMs) in [Compute Engine](#) with microservices in [Google Kubernetes Engine](#) (GKE) to run SmartDaily for higher software update efficiency, and adopt information security solutions like [Security Command Center](#) to be compliant with global standards, including ISO27001.

“Over the past seven years, CloudMile has been constantly providing great help by sharing their rich experience in cloud computing and information security. It’s like having a reliable consultant always by our side,” notes Chang.

Leveraging the cloud infrastructure of Google Cloud has greatly improved the stability of SmartDaily’s smart community service platform, which has never experienced any major downtime since the migration. As the platform incorporates a large number of systems and features, the SmartDaily team needs to release an app update every day. When SmartDaily was deployed in Compute Engine, it required more than 10 minutes to manually set up a rolling update. Now, since GKE supports automatic deployment, SmartDaily is able to release an app update with just one click and invest more resources in development work.

On top of that, SmartDaily uses [Cloud SQL](#) to host the relational databases of all its services. Chang says that the fully-managed services of Cloud SQL greatly simplify database settings and management for SmartDaily, which enables the team to focus on application R&D.

“Moving our smart service platform from on-prem to Google Cloud has benefited us in many aspects from security, service stability to productivity,” he adds. “Having seen the excellent results, we migrated our operational systems to Google Cloud in 2024 as well to further streamline our management.”

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Boosting data analytics efficiency with a centralized data warehouse in BigQuery

To continuously improve its services and operations, SmartDaily relies on data analytics to gain insights. Previously, the company did not have a centralized data warehouse. Whenever there was an analytics demand, its engineers had to pull out relevant data from different sources to create a dataset specific to the request. This method was not only resource-consuming, but also limited the team’s analytics capability.

In 2021, SmartDaily decided to form a data team and build a centralized data warehouse in [BigQuery](#) aggregating all its user behavior and device data in Cloud SQL and [Google Analytics](#). Now, its data team can immediately retrieve the data needed when an analytics demand comes in without creating new datasets, which could take up to one week. This way, SmartDaily’s different departments are able to run more analytics in a shorter period of time and see visualized results directly in [Looker Studio](#).

“Having a centralized data warehouse in BigQuery has significantly broadened and accelerated our data analytics, enabling us to optimize our operations and products more effectively and timely. For example, we’re now able to understand our customers better, improve users’ usage intention and stay time, and increase the number of active users by nearly 20%,” says Chang.

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Providing faster customer service with predictive analytics

Serving more than two million users, SmartDaily's customer service team receives numerous calls every day. Since the company provides diverse services, its customer service agents need to have extensive knowledge across different industries or know where to find information required in a short time to answer users' questions. To enhance the quality of its customer service, SmartDaily in early 2024 built a machine learning (ML) model in BigQuery, and fed it with frequently asked questions tagged by [Vertex AI](#) and industry knowledge summarized by [Gemini 1.5 Pro](#). Using the predicted questions and answers generated by the ML model, SmartDaily's customer service team can respond to customers' requests much more quickly, shortening the time required to process one call from 15 to seven minutes.

"Our users include many elderly people, who heavily rely on phone calls and human agents for customer service, so we need to first improve our customer service efficiency through ML, instead of directly developing a chatbot," explains Chang. "With the AI and ML capabilities of Google Cloud, we're able to provide faster customer service while ensuring that our agents have all the information they need at hand."

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Improving service quality with AI

Next, SmartDaily will develop virtual customer service agents using [Contact Center AI](#), [Dialogflow](#), [Text-to-Speech AI](#), and [Speech-to-Text AI](#) to support its call center by phone and online. To improve the quality of its services, the company also plans to build a ML model in BigQuery that can give work guidance to its partner technicians or service providers based on their past work history, so that they will have a better idea about how they can provide better services.

Chang says, “As AI technology becomes more mature and accessible, our next step is to elevate our service quality by integrating AI-powered tools into our work process. The industry-leading AI capabilities of Google Cloud can definitely help us reach this goal and create better community living experiences for our users.”

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Founded in 1997, KNST CO., LTD. is one of the largest smart community service providers in Taiwan, serving nearly 9,000 communities and more than a million families. Dedicated to enabling convenient, pleasant community living experiences for its users, KNST CO., LTD. in 2016 launched **SmartDaily**, a smart community service platform offering wide-ranging AIoT-powered community management and domestic services.

Industry: Technology

Location: Taiwan

Products: [BigQuery](#), [Vertex AI](#), [Cloud SQL](#), [Compute Engine](#), [Google Kubernetes Engine](#), [Looker Studio](#), [Security Command Center](#)

About Google Cloud Partner- [CloudMile](#)

As one of the leading AI and cloud service providers in Taiwan, [CloudMile](#) empowers businesses to accelerate digital transformation through cloud technology and machine learning. Currently holding more than 200 professional certifications and having served 900+ enterprise customers, CloudMile is a Google Cloud partner specializing in infrastructure, machine learning, data analytics, cloud security, and work transformation.

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